

GRONINGEN EELDE, Netherlands

WMO: 062800

Lat: 53.1236N

Lon: 6.5847E

Elev: 17

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.4	22.0	12.8	10.6	20.4	17.2	13.1	23.3	32.2	47.2	29.4	47.8	7.6	80	0.593

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.9	83.8	68.2	79.3	66.7	75.4	64.8	70.0	79.7	68.1	77.1	66.2	73.5	8.4	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.7	98.5	74.2	64.9	92.1	71.6	63.2	86.9	69.6	34.0	79.6	32.3	77.3	30.9	73.9	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
25.3	22.0	19.1	DB	12.7	90.9	6.7	2.6	7.9	92.8	4.0	94.3	0.2	95.7	-4.7	97.6
			WB	12.2	74.1	6.7	1.9	7.4	75.4	3.5	76.5	-0.3	77.5	-5.1	78.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	49.6	36.7	37.5	41.8	48.3	54.3	59.3	63.6	63.2	57.7	50.9	43.5	38.1	(6)
(7)		DBStd	11.50	7.51	7.21	6.09	6.62	6.40	5.58	5.63	5.25	4.89	6.04	6.03	7.66	(7)
(8)		HDD50	1807	414	350	261	111	27	1	0	0	1	62	208	372	(8)
(9)		HDD65	5737	878	769	720	503	336	189	97	100	226	438	646	835	(9)
(10)		CDD50	1672	1	1	6	59	161	279	422	408	232	90	11	2	(10)
(11)		CDD65	127	0	0	0	0	6	17	54	43	7	0	0	0	(11)
(12)		CDH74	1144	0	0	0	14	69	183	475	347	55	1	0	0	(12)
(13)		CDH80	312	0	0	0	1	7	46	144	106	8	0	0	0	(13)
(14)	Wind	WSAvg	9.1	11.1	10.6	10.5	9.1	8.7	8.3	7.8	7.5	7.8	8.8	9.2	10.4	(14)
(15)	Precipitation	PrecAvg	31.1	2.6	1.9	2.1	1.9	2.2	2.8	3.1	2.7	3.0	2.7	3.1	3.1	(15)
(16)		PrecMax	43.0	5.7	5.2	5.3	4.0	5.7	5.7	6.3	6.2	8.7	7.1	5.1	6.6	(16)
(17)		PrecMin	21.0	0.2	0.2	0.3	0.2	0.6	0.9	0.6	0.2	0.6	0.6	0.8	0.7	(17)
(18)		PrecStd	5.1	1.3	1.1	1.1	1.0	1.1	1.2	1.4	1.4	1.6	1.3	1.1	1.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.8	55.6	63.3	75.8	80.5	85.7	90.1	88.3	80.4	70.5	59.9	55.2	(19)
(20)			MCWB	51.3	48.9	52.8	59.5	64.6	69.5	69.6	69.2	67.1	62.4	56.4	53.1	(20)
(21)		2%	DB	50.8	51.7	57.8	69.4	75.3	79.3	84.0	82.6	74.1	65.0	56.8	52.8	(21)
(22)			MCWB	48.9	48.3	50.7	56.0	61.0	66.4	68.8	68.2	64.8	60.0	54.8	50.9	(22)
(23)		5%	DB	48.5	49.3	53.8	64.3	71.0	74.2	79.2	77.1	69.7	61.9	54.4	50.6	(23)
(24)			MCWB	46.8	46.8	48.7	54.5	60.0	64.3	67.3	66.8	62.4	58.0	52.2	48.9	(24)
(25)		10%	DB	46.5	47.2	51.0	59.9	66.5	70.0	74.8	73.0	66.2	59.5	52.1	48.1	(25)
(26)			MCWB	44.8	45.1	47.3	52.6	58.1	62.0	65.5	64.6	60.2	56.6	50.1	46.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.6	51.1	54.8	60.9	67.5	71.2	73.1	72.5	68.4	63.2	57.4	53.6	(27)
(28)			MCD B	53.2	53.6	59.9	71.8	77.2	81.6	83.9	82.7	77.7	68.6	58.5	55.0	(28)
(29)		2%	WB	49.2	48.8	51.8	57.7	63.7	67.8	70.3	69.3	65.5	60.6	55.0	51.3	(29)
(30)			MCD B	50.6	50.7	55.7	66.4	72.0	76.5	80.0	78.2	72.5	64.2	56.8	52.5	(30)
(31)		5%	WB	47.0	47.0	49.7	55.5	61.0	65.3	68.3	67.3	63.2	58.7	52.4	49.0	(31)
(32)			MCD B	48.3	49.0	53.1	62.7	68.2	72.3	77.5	75.6	68.5	61.4	54.1	50.5	(32)
(33)		10%	WB	45.0	45.2	47.5	53.2	58.5	62.9	66.3	65.5	61.4	57.0	50.2	46.6	(33)
(34)			MCD B	46.4	47.2	50.6	59.1	65.5	68.9	73.7	71.9	65.1	59.3	51.8	47.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.1	9.9	13.0	16.8	17.1	16.9	16.9	17.1	15.4	12.8	9.3	8.2	(35)
(36)			MCDBR	8.4	11.4	18.4	24.2	24.0	23.9	24.2	23.4	20.0	14.6	10.1	8.3	(36)
(37)			MCWBR	7.8	9.0	12.4	13.6	12.9	13.0	11.4	11.3	11.6	10.1	8.7	7.8	(37)
(38)		5% WB	MCDBR	8.3	10.3	16.3	21.4	21.3	21.5	21.5	20.6	18.1	13.9	9.8	8.5	(38)
(39)			MCWBR	8.0	8.7	11.7	12.9	12.6	12.6	11.3	11.0	11.1	10.2	8.8	8.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.327	0.346	0.375	0.406	0.405	0.403	0.415	0.408	0.388	0.361	0.340	0.320	(40)	
(41)		taud	2.320	2.331	2.268	2.232	2.272	2.295	2.275	2.310	2.360	2.400	2.349	2.310	(41)	
(42)		Ebn at Noon	203	238	254	261	268	269	263	258	248	230	196	181	(42)	
(43)		Edh at Noon	19	26	33	39	40	39	39	36	30	24	19	16	(43)	
(44)	All-Sky Solar Radiation	RadAvg	194	402	781	1292	1596	1681	1598	1331	951	538	236	150	(44)	
(45)		RadStd	20	65	90	130	135	123	190	108	100	54	23	16	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46)
(47)	Regional (52 neighbors)	(m) N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page